

Urban Heat Islands

Student sheet

Name: _____

Date: _____

1. What I see on the screen

A single chart, with one **coloured line per box**, labelled with the box's name. Below it, the ΔT card, which shows how much the warmest box differs from the coolest and names both of them. You save with **Export data**, top right. **There is no green button here, no Back, no Finish, no Pause**, and in the device list the value on the right stays —°C: the numbers are read from the card and from the chart.

The app's own verdict — “similar temperatures” below 2°C, “moderate difference” above 2, “strong effect” above 5 — **has fixed thresholds and knows nothing about our boxes**. We compare ΔT with **our own spread**. I wait **five minutes** at each location and I do not hold the box in my hand.

2. What I do on my own

Weather: _____ **Height used:** _____ cm **How many boxes are in the list?** _____

First I predict. I do not change the prediction after I measure. **1.** The warmest location will be _____, the coolest _____. **2.** The difference between them will be _____ °C. **3.** Two boxes placed *side by side* will show numbers that differ by _____ °C.

Co-location. All boxes side by side, on the same table, out of the sun, **five minutes**. Only then do I read.

#	Box name, as shown on screen	T (°C)	Where I take it
1			
2			
3			
4			

SPREAD = largest – smallest = _____ – _____ = _____ °C

Now I measure in the field. The same height at all boxes, the same shading, five minutes, the time noted.

Location (surface)	Box	Time	T (°C)	RH (%)

ΔT = warmest _____ – coolest _____ = _____ °C Warmest box: _____

If I repeat the route (project route), I record for each outing the time, the spread and ΔT . **The spread is measured again every time.**

Morning: _____ / _____ / _____ Noon: _____ / _____ / _____ End of day: _____ / _____ / _____

3. The conclusion I write

My ΔT : _____ °C. Our spread: _____ °C. So the difference ☐ is detectable ☐ **is not detectable** with our setup. *“Not detectable” does not mean “does not exist”.*

I predicted that, and the data show that _____

Our spread was _____

The warmest location was the warmest because _____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.
